



BACHELOR OF INFORMATION TECHNOLOGY (HONS)

FINAL EXAMINATION DECEMBER 2025

Course : EC3114 (Software Modelling and Analysis) Time: 8.00 am – 11.00 am
(3 hours)
Lecturer : Mr. Izrulfizal Saufihamizal bin Ibrahim Date : 18 February 2026

Instructions:

Answer **ALL** questions in the Answer Booklet provided.

The maximum number of marks is 100.

This marking scheme consists of 2 printed pages.
(excluding front cover)

DO NOT OPEN THIS BOOKLET UNTIL YOU ARE TOLD TO DO SO

1. a. Analyze the techniques used for requirement elicitation. Which technique would be most effective for a system that interacts with a wide range of users? Justify your answer. **(8 MARKS)**
b. Differentiate between functional, non-functional requirements, and constraints. Provide examples of each from a hospital management system. **(5 MARKS)**
c. Discuss the deliverables of each phase of the SDLC. Which phase do you consider most critical for success and why? **(12 MARKS)**
2. a. One of the tools to support modelling in functional requirements is Use Case Diagram. Explain how Use Case Diagrams support functional requirement analysis. **(5 MARKS)**
b. Define a sequence diagram and explain how it differs from a collaboration diagram. **(5 MARKS)**
c. Explain the steps in the requirements gathering process and discuss why collecting accurate requirements early in software development is essential. **(10 MARKS)**
d. Discuss the role of screen prototypes in validating system design with stakeholders. **(5 MARKS)**
3. You are tasked with designing a class diagram for a simple Online Shopping System. The system should include functionalities for managing products, customers, and orders. Your class diagram should include at least three classes, with attributes, methods, and relationships clearly defined. Provide explanations for your design choices and any assumptions you make. **(25 MARKS)**

4. Create an activity diagram to model the process of online food ordering on a restaurant platform. The process should include activities such as browsing the menu, selecting items, adding them to the cart, placing an order, payment processing, and order confirmation. Clearly label all activities, decision points, and control flows. Additionally, explain how this activity diagram can help in optimizing the platform's efficiency and user satisfaction. **(25 MARKS)**

-END OF QUESTION PAPER-